Status	Finished
Started	Tuesday, 18 August 2026, 9:29 AM
Completed	Thursday, 20 August 2026, 1:03 PM
Duration	2 days 3 hours
Marks	153.00/155.00
Grade	**9.87** out of 10.00 (**98.71**%)

Question 1

Correct

Mark 35.00 out of 35.00

What is the single-precision format of the floating-point number: 19.6875?

Solution:

1. **Sign bit:** 0
2. **Convert the integer to binary**:

   Integer:

   2.1. 19/2 → 1

   2.2. 9 /2 → 1

   2.3. 4 /2 → 0

   2.4. 2 /2 → 0

   2.5. 1 /2 → 1

   Then the binary of the **integer 19:** 1 0 0 1 1

   Fraction:

   2.6. 0.6875x2 → 1

   2.7. 0.375 x2 → 0

   2.8. 0.75 x2 → 1

   2.9. 0.5 x2 → 1

   Then the binary of the **fraction 0.6875:** 1 0 1 1
3. **Normalize**: 1.00111011 × 2^ 4
4. **Exponent**: 127 + 4 = 131 → 10000011
5. **Mantissa**: 00111011000000000000000
6. **Final:** 0 | 10000011 | 00111011000000000000000

Question 2

Correct

Mark 36.00 out of 36.00

What is the single-precision format of the floating-point number: -1/3?
Solution:

1. **Sign bit:** 1
2. **Convert the integer to binary**:
   Integer: 0 → 0
   Fraction:
   2.1. 0.333... (show at least 3 digits with ...) x2 → 0
   2.2. 0.666... x2 → 1
   2.3. 0.333... x2 → 0
   It's repeating, then the binary of the **fraction of -1/3:** 0 1 0 1 **...**
3. **Normalize**: 1.01010101... (show at least 8 digits of fraction with ...)× 2^ -2
4. **Exponent**: 127 + -2 = 125 → 01111101
5. **Mantissa**: 01010101010101010101011

   Note: IEEE 754 Rounding Rules
   1. Guard bit (24th bit): Determines rounding.
   2. Round bit (25th bit) & Sticky bit (OR of bits 25–29): Used for tie-breaking.
   Rules:
   1. If guard bit = 0 → Truncate (no rounding).
   2. If guard bit = 1 → Check further:
      2.1. If any bit after (round/sticky) = 1 → Round up (add 1 to LSB and then calculate as the binary number operation).
      2.2. If all remaining bits = 0 → Round to even (make LSB 0 if even).
6. **Final:** 1 | 01111101 | 01010101010101010101011
7. **Split into 4-bit groups:** if it is a 32-bit format, use only the first row
and in the second row put 'none', but if it is a 64-bit format, use both.

Row1:	1011	1110	1010	1010	1010	1010	1010	1011
Row2:	none	none	none	none	none	none	none	none

Question 3

Correct

Mark 33.00 out of 33.00

What is the double-precision format of the floating-point number: -3.75?
Solution:

1. **Sign bit:** 1
2. **Convert the integer to binary**:
   Integer: 3 → 11
   Fraction:
   2.1. 0.75 x2 → 1
   2.2. 0.5 x2 → 1
   It's repeating, then the binary of the **fraction 0.75:** 1 1
3. **Normalize**: 1.111 × 2^ 1
4. **Exponent**: 1023 (bias value) + 1 = 1024 → 10000000000
5. **Mantissa**: 111 (put only the value you got from the normalization and then answer the next question).
   Question: How many zeros do we need after the last bit of the normalization process? 49
6. **Split into 4-bit groups for the final:** if it is a 32-bit format, use only the first row, and the second row just put 'none', but if it is a 64-bit format, use both.

Row1: 1100 0000 0000 1110 0000 0000 0000 0000
Row2: 0000 0000 0000 0000 0000 0000 0000 0000

Question 4

Correct

Mark 28.00 out of 28.00

What is the double-precision format of the floating-point number: -0.1?

Solution:

1. **Sign bit:** 1

2. **Convert the integer to binary**:

   Integer: 0 → 0

   Fraction:

   2.1. 0.1 x2 → 0

   2.2. 0.2 x2 → 0

   2.3. 0.4 x2 → 0

   2.4. 0.8 x2 → 1

   2.5. 0.6 x2 → 1

   It's repeating, then the binary of the **fraction of -0.1:** 0 0 0 1 1 0 0 1 1 0 ...

3. **Normalize**: 1.1001100110011001... (show at least 16 digits of fraction with ...)× 2^ -4

4. **Exponent**: 1023 + -4 = 1019 → 01111111011

5. **Mantissa**: What are the last 4 digits with LSB? ... 1010

   Note: IEEE 754 Rounding Rules

   1. Guard bit (24th bit): Determines rounding.
   2. Round bit (25th bit) & Sticky bit (OR of bits 25–29): Used for tie-breaking.

   Rules:

   1. If guard bit = 0 → Truncate (no rounding).
   2. If guard bit = 1 → Check further:
      2.1. If any bit after (round/sticky) = 1 → Round up (add 1 to LSB and then calculate as the binary number operation).
      2.2. If all remaining bits = 0 → Round to even (make LSB 0 if even).

Question 5

Partially correct

Mark 21.00 out of 23.00

Instruction A:
The `ID/EX` pipeline register contains the following 32-bit instruction:
`000000 01010 01011 01100 00000 100000`

Instruction B:
The `ID/EX` pipeline register contains the following 32-bit instruction:
`100011 01010 01011 0000 0000 0000 0101`

Identify the Instruction Type (a MUX task): For both **Instruction A** and **Instruction B**, identify whether they are an R-Type or an I-Type instruction. Explain your reasoning based on the specific bits of the instruction.

1. **Instruction A** is R-Type because the opcode (bits 31–26) is 000000

   MUX Output: The MUX will select the second input from rt , which is the data from the register specified in the rt field, from bit number 12 to bit number 16 , (if LSB is named bit number 0 and MSB is named bit number 31), which is 0 1 0 1 1 . Therefore, the ALU will receive the value stored in `$t3`.
2. **Instruction B** is I-Type because the first 6 bits (at MSB) are 1 0 0 0 1 1 which is the opcode for the `lw` (load word) instruction, a classic I-Type instruction.

   MUX Output: The MUX will select the sign-extended immediate value from the instruction's immediate field (the last 16 bits, which is decimal 5). The MUX will pass this value to the ALU. This will be used to compute the memory address by adding the offset to the base address in `$t2`.